1. Basic MySQL Questions

1. What is MySQL?
Answer: MySQL is an open-source Relational Database Management System (RDBMS).
2. What is a Database?
Answer: A database is an organized collection of related data.
3. What is DBMS?
Answer: DBMS stands for Database Management System. It is software used to create, manage and control databases.
4. What is RDBMS?
Answer: RDBMS stands for Relational Database Management System. It stores data in related tables.
5. What is SQL?
Answer: SQL stands for Structured Query Language. It is used to communicate with databases.
6. What is a Table?
Answer: A table is a collection of data organized into rows and columns.
7. What is a Row?
Answer: A row represents a single record in a table.
8. What is a Column?
Answer: A column represents a field or attribute in a table.
9. What is a Primary Key?
Answer: A Primary Key uniquely identifies each record in a table.
10. What is a Foreign Key?
Answer: A Foreign Key is a column that creates a relationship between two tables.
11. What is NULL?
Answer: NULL represents a missing or unknown value in a database field.
12. What is a Constraint?
Answer: A constraint is a rule applied to table columns to maintain data accuracy and integrity.

2. SQL Command Questions

13. What is DDL?
Answer: DDL stands for Data Definition Language. It defines database structures.
14. What is DML?
Answer: DML stands for Data Manipulation Language. It is used to insert, update and delete data.
15. What is DQL?
Answer: DQL stands for Data Query Language. It is used to retrieve data from a database.
16. What is DCL?
Answer: DCL stands for Data Control Language. It controls database access permissions.
17. What is TCL?
Answer: TCL stands for Transaction Control Language. It manages database transactions.
18. What is the use of CREATE command?
Answer: It is used to create a database or table.
19. What is the use of INSERT command?
Answer: It is used to insert new records into a table.

20. What is the use of SELECT command?
Answer: It is used to retrieve data from a table.
21. What is the use of UPDATE command?
Answer: It is used to modify existing records.
22. What is the use of DELETE command?
Answer: It is used to delete records from a table.
23. What is the use of DROP command?
Answer: It is used to permanently remove a database or table.
24. What is the use of TRUNCATE command?
Answer: It removes all records from a table while keeping its structure.

3. SQL Clauses and Functions

25. What is WHERE?
Answer: WHERE is used to filter records based on specific conditions.
26. What is ORDER BY?
Answer: ORDER BY is used to sort records in ascending or descending order.
27. What is GROUP BY?
Answer: GROUP BY is used to group rows with the same values.
28. What is HAVING?
Answer: HAVING is used to filter grouped records.
29. What is JOIN?
Answer: JOIN is used to combine related data from two or more tables.
30. What is INNER JOIN?
Answer: INNER JOIN returns matching records from both tables.
31. What is LEFT JOIN?
Answer: LEFT JOIN returns all records from the left table and matching records from the right table.
32. What is RIGHT JOIN?
Answer: RIGHT JOIN returns all records from the right table and matching records from the left table.
33. What is COUNT()?
Answer: COUNT() returns the total number of records.
34. What is SUM()?
Answer: SUM() calculates the total of numeric values.
35. What is AVG()?
Answer: AVG() calculates the average value.
36. What is MAX()?
Answer: MAX() returns the highest value.
37. What is MIN()?
Answer: MIN() returns the lowest value.
38. What is LIKE operator?
Answer: LIKE is used to search for a specific pattern in a column.
39. What is DISTINCT?
Answer: DISTINCT is used to retrieve unique values and remove duplicates from the result.

4. Advanced MySQL Questions

40. What is a Subquery?
Answer: A Subquery is a query written inside another SQL query.
41. What is a View?
Answer: A View is a virtual table based on the result of an SQL query.
42. What is an Index?
Answer: An Index helps retrieve data from a database faster.
43. What is a Stored Procedure?
Answer: A Stored Procedure is a group of SQL statements stored in a database for reuse.
44. What is a Trigger?
Answer: A Trigger is a database object that executes automatically when a specified database event occurs.
45. What is a Window Function?
Answer: A Window Function performs calculations across related rows without combining them into a single row.
46. What is a Transaction?
Answer: A Transaction is a group of database operations executed as a single logical unit.
47. What is COMMIT?
Answer: COMMIT permanently saves transaction changes.
48. What is ROLLBACK?
Answer: ROLLBACK cancels uncommitted transaction changes.
49. What is AUTO_INCREMENT?
Answer: AUTO_INCREMENT automatically generates a new numeric value for a column.
50. What is UNIQUE Constraint?
Answer: UNIQUE Constraint prevents duplicate values in a column or combination of columns.

5. MySQL Practical Questions

No.	Practical Question	SQL Command
1	Create a database.	CREATE DATABASE College;
2	Select a database.	USE College;
3	Create a table.	CREATE TABLE Student (ID INT, Name VARCHAR(50));
4	Insert a record.	INSERT INTO Student VALUES (1,'Rahim');
5	Display all records.	SELECT * FROM Student;
6	Display a specific record.	SELECT * FROM Student WHERE ID=1;
7	Update a record.	UPDATE Student SET Name='Karim' WHERE ID=1;
8	Delete a record.	DELETE FROM Student WHERE ID=1;
9	Sort records.	SELECT * FROM Student ORDER BY ID DESC;
10	Count total records.	SELECT COUNT(*) FROM Student;
11	Display unique names.	SELECT DISTINCT Name FROM Student;
12	Delete a table.	DROP TABLE Student;

6. Database Administration and Practical Viva

51. What is the default port number of MySQL?
Answer: 3306.
52. What is the default administrator username in MySQL?
Answer: Usually root.
53. What is Database Backup?
Answer: Database Backup is the process of creating a copy of database data for future use.
54. What is Database Restore?
Answer: Database Restore is the process of recovering a database from a backup.
55. What is Database Normalization?
Answer: Normalization is the process of organizing data to reduce redundancy and improve data integrity.
56. What is an ER Diagram?
Answer: An ER Diagram is a graphical representation of entities and their relationships.
57. What is the use of SHOW DATABASES command?
Answer: It displays all available databases.
58. What is the use of DESCRIBE command?
Answer: It displays the structure of a table.
59. How do you write comments in MySQL?
Answer: Comments can be written using -- or /* ... */.
60. What is the difference between CHAR and VARCHAR?
Answer: CHAR stores fixed-length strings, while VARCHAR stores variable-length strings.

Short Questions and answers on Design and Model Database

1. Basic Database Design Questions
1. What is Database Design?
Answer: Database Design is the process of planning and organizing data into tables, columns, and relationships.
2. What is Database Modeling?
Answer: Database Modeling is the process of creating a visual representation of a database structure and its relationships.
3. Why is Database Design important?
Answer: It helps to reduce data redundancy, maintain data accuracy, and improve database performance.
4. What is Data Modeling?
Answer: Data Modeling is the process of defining how data is stored, organized, and related.
5. What is an Entity?
Answer: An Entity is a real-world object or concept about which data is stored.
6. What is an Attribute?
Answer: An Attribute is a property or characteristic of an entity.

7. What is a Relationship?
Answer: A Relationship represents an association between two or more entities.
8. What is a Database Schema?
Answer: A Database Schema is the logical structure or design of a database.
9. What is a Database Instance?
Answer: A Database Instance is the actual data stored in a database at a particular time.
10. What is Data Redundancy?
Answer: Data Redundancy means storing the same data unnecessarily in multiple places.

2. ER Diagram and Database Modeling

11. What is an ER Diagram?
Answer: An ER Diagram (Entity Relationship Diagram) is a graphical representation of entities, attributes, and relationships.
12. What is an Entity Set?
Answer: An Entity Set is a collection of similar entities.
13. What is a Primary Key?
Answer: A Primary Key uniquely identifies each record in a table.
14. What is a Foreign Key?
Answer: A Foreign Key is a column that creates a relationship between two tables.
15. What is a Candidate Key?
Answer: A Candidate Key is an attribute or set of attributes that can uniquely identify a record.
16. What is a Composite Key?
Answer: A Composite Key is a key formed by two or more columns.
17. What is a Super Key?
Answer: A Super Key is a set of one or more attributes that uniquely identifies a record.
18. What is Cardinality?
Answer: Cardinality defines the number of relationships between entities.
19. What is One-to-One Relationship?
Answer: It is a relationship where one record in a table is associated with one record in another table.
20. What is One-to-Many Relationship?
Answer: It is a relationship where one record in a table is associated with multiple records in another table.
21. What is Many-to-Many Relationship?
Answer: It is a relationship where multiple records in one table are associated with multiple records in another table.

3. Database Normalization Questions

22. What is Normalization?
Answer: Normalization is the process of organizing data to reduce redundancy and improve data integrity.
23. What is First Normal Form (1NF)?
Answer: 1NF requires each column to contain atomic values without repeating groups.
24. What is Second Normal Form (2NF)?
Answer: 2NF requires 1NF and no partial dependency of non-key attributes on a candidate key.

25. What is Third Normal Form (3NF)?
Answer: 3NF requires 2NF and no transitive dependency of non-key attributes on candidate keys.
26. What is Denormalization?
Answer: Denormalization is the process of combining data to improve read performance, sometimes by introducing redundancy.
27. What is Functional Dependency?
Answer: Functional Dependency means one attribute determines the value of another attribute.
28. What is Data Integrity?
Answer: Data Integrity means maintaining the accuracy, consistency, and reliability of data.

4. Database Design Concepts

29. What is a Data Type?
Answer: A Data Type defines the kind of data that can be stored in a column.
30. What is VARCHAR?
Answer: VARCHAR is a variable-length character data type.
31. What is INT?
Answer: INT is a data type used to store integer values.
32. What is DATE?
Answer: DATE is a data type used to store date values.
33. What is a Constraint?
Answer: A Constraint is a rule applied to a database column or table to maintain data accuracy.
34. What is NOT NULL Constraint?
Answer: NOT NULL prevents a column from storing NULL values.
35. What is UNIQUE Constraint?
Answer: UNIQUE prevents duplicate values in a column or combination of columns.
36. What is DEFAULT Constraint?
Answer: DEFAULT assigns a predefined value when no value is provided.
37. What is CHECK Constraint?
Answer: CHECK ensures that values satisfy a specified condition.

5. Database Design and Modeling Tools

38. What is MySQL Workbench?
Answer: MySQL Workbench is a visual tool used for database design, SQL development, and database administration.
39. What is phpMyAdmin?
Answer: phpMyAdmin is a web-based tool used to manage MySQL and MariaDB databases.
40. What is UML?
Answer: UML stands for Unified Modeling Language. It is used to visually represent software systems and their structures.
41. What is a Logical Data Model?
Answer: It describes entities, attributes, keys, and relationships without focusing on physical storage.
42. What is a Physical Data Model?
Answer: It describes how the database is implemented, including tables, columns, data types, indexes, and constraints.

43. What is a Conceptual Data Model?
Answer: It provides a high-level overview of entities and their relationships.

6. Practical Database Design Questions

44. What is the first step in database design?
Answer: Collecting and analyzing user requirements.
45. How do you create a database in MySQL?

```
CREATE DATABASE College;
```

46. How do you create a Student table?

```
CREATE TABLE Student (
   Student_ID INT PRIMARY KEY,
   Student_Name VARCHAR(50),
   Department VARCHAR(30),
   Age INT
);
```

47. How do you create a Foreign Key?

```
CREATE TABLE Department (
   Dept_ID INT PRIMARY KEY,
   Dept_Name VARCHAR(50)
);

CREATE TABLE Student (
   Student_ID INT PRIMARY KEY,
   Student_Name VARCHAR(50),
   Dept_ID INT,
   FOREIGN KEY (Dept_ID)
   REFERENCES Department(Dept_ID)
);
```

48. How do you display table structure?

```
DESCRIBE Student;
```

49. How do you add a new column to a table?

```
ALTER TABLE Student
ADD Email VARCHAR(100);
```

50. How do you create an Index?

```
CREATE INDEX idx_student_name
ON Student(Student_Name);
```

Short Questions and Answers on Set Up Database Environment

A. Basic Concepts

1. What is a database environment?
A database environment is a system consisting of hardware, software, database, users, and procedures used to manage data.
2. What is database setup?
Database setup is the process of installing and configuring database software and preparing it for use.
3. What is a DBMS?
DBMS stands for Database Management System. It is software used to create, store, manage, and retrieve data.
4. Give examples of DBMS software.
MySQL, Oracle, Microsoft SQL Server, PostgreSQL, and SQLite.
5. What is MySQL?
MySQL is an open-source relational database management system.
6. What is a database server?
A database server is a computer or software system that stores and manages databases.
7. What is a database client?
A database client is an application or tool used to connect to and interact with a database server.
8. What is SQL?
SQL stands for Structured Query Language. It is used to manage relational databases.
9. What is a database administrator (DBA)?
A DBA is responsible for installing, configuring, securing, and maintaining databases.
10. What is a relational database?
A relational database stores data in tables consisting of rows and columns.

B. Database Installation and Configuration

11. What is database software installation?
It is the process of installing DBMS software on a computer or server.
12. What is a database configuration?
Database configuration means setting parameters such as port number, storage location, memory, and security options.
13. What is the default port number of MySQL?
The default port number of MySQL is 3306.
14. What is the default port number of PostgreSQL?
The default port number of PostgreSQL is 5432.
15. What is XAMPP?
XAMPP is a software package that includes Apache, MariaDB, PHP, and Perl.
16. What is phpMyAdmin?
phpMyAdmin is a web-based tool used to manage MySQL or MariaDB databases.
17. What is a database connection?
A database connection is a link between a client application and a database server.

18. What information is required to connect to a database?
Hostname, port number, username, password, and database name.
19. What is a database service?
A database service is a background process that runs and manages the database server.
20. Why should we configure a database server?
To ensure proper performance, security, connectivity, and reliability.

C. Database Creation and Management

21. Which SQL command creates a database?
CREATE DATABASE
22. Write the SQL command to create a database.
CREATE DATABASE College;
23. Which SQL command selects a database in MySQL?
USE
24. Write the command to select a database.
USE College;
25. Which SQL command creates a table?
CREATE TABLE
26. Write a command to create a Student table.

```
CREATE TABLE Student (
   ID INT PRIMARY KEY,
   Name VARCHAR(50),
   Age INT
);
```

27. Which SQL command displays available databases?
SHOW DATABASES;
28. Which SQL command displays tables?
SHOW TABLES;
29. What is a primary key?
A primary key uniquely identifies each record in a table.
30. What is a foreign key?
A foreign key creates a relationship between two tables.

D. Security and Maintenance

31. Why is database security important?
It protects data from unauthorized access, modification, and deletion.
32. What is a database user?
A database user is an account that has permission to access a database.
33. What is database authentication?
Authentication is the process of verifying a user's identity.
34. What is database authorization?
Authorization determines what actions an authenticated user can perform.
35. What is database backup?
Backup is the process of creating a copy of database data for recovery.
36. What is database restoration?
Restoration is the process of recovering database data from a backup.

37. What is database monitoring?
Database monitoring is the process of checking database performance, availability, and errors.
38. What is database optimization?
Database optimization improves database performance and query execution speed.
39. Why should database passwords be strong?
Strong passwords help prevent unauthorized access.
40. What is a firewall?
A firewall is a security system that controls incoming and outgoing network traffic.

E. Practical Questions

41. How do you check the MySQL server version?

```
SELECT VERSION();
```

42. How do you check the current database?

```
SELECT DATABASE();
```

43. How do you display the current MySQL user?

```
SELECT USER();
```

44. How do you create a database user?

```
CREATE USER 'student'@'localhost'
IDENTIFIED BY 'StrongPass123!';
```

45. How do you grant database privileges?

```
GRANT ALL PRIVILEGES ON College.*
TO 'student'@'localhost';
```

46. How do you display database tables?

```
SHOW TABLES;
```

47. How do you check the structure of a table?

```
DESCRIBE Student;
```

48. How do you insert data into a table?

```
INSERT INTO Student (ID, Name, Age)
VALUES (1, 'Rahim', 20);
```

49. How do you retrieve data from a table?

```
SELECT * FROM Student;
```

50. How do you delete a database?

```
DROP DATABASE College;
```

Short questions and answers on Create and Manage Database Objects

A. Basic Concepts

1. What is a database object?

A database object is a component used to store, manage, or process data in a database.

2. What are the common database objects?

Tables, views, indexes, stored procedures, functions, and triggers.

3. What is a table?

A table is a database object that stores data in rows and columns.

4. What is a row?

A row is a single record in a table.

5. What is a column?

A column represents a specific attribute or field in a table.

6. What is a view?

A view is a virtual table based on an SQL query.

7. What is an index?

An index is a database object that improves data retrieval speed.

8. What is a stored procedure?

A stored procedure is a group of SQL statements stored in a database.

9. What is a database function?

A function is a stored program that performs a task and returns a value.

10. What is a trigger?

A trigger is a stored program that automatically executes when a specified database event occurs.

B. Creating Database Objects

11. Which SQL command creates a table?

CREATE TABLE

12. Write SQL syntax to create a table.

```
CREATE TABLE Student (
   ID INT PRIMARY KEY,
   Name VARCHAR(50),
   Age INT
);
```

13. Which command creates a view?

CREATE VIEW

14. Write SQL syntax to create a view.

```
CREATE VIEW Student_View AS
SELECT ID, Name FROM Student;
```

15. Which command creates an index?

CREATE INDEX

16. Write SQL syntax to create an index.

```
CREATE INDEX idx_name
ON Student(Name);
```

17. Which command creates a stored procedure?
CREATE PROCEDURE
18. Write a simple stored procedure.

```
DELIMITER //

CREATE PROCEDURE GetStudents()
BEGIN
   SELECT * FROM Student;
END //

DELIMITER ;
```

19. Which command creates a database function?
CREATE FUNCTION
20. What is the purpose of CREATE TRIGGER?
It creates a trigger that automatically executes after or before a specified database event.

C. Managing Database Objects

21. Which command displays all tables?
SHOW TABLES;
22. Which command displays table structure?
DESCRIBE Student;
23. Which command modifies a table structure?
ALTER TABLE
24. How do you add a new column to a table?

```
ALTER TABLE Student
ADD Email VARCHAR(100);
```

25. How do you modify a column data type?

```
ALTER TABLE Student
MODIFY Age SMALLINT;
```

26. How do you rename a table?

```
RENAME TABLE Student
TO Students;
```

27. Which command deletes a table?
DROP TABLE
28. Which command removes all records from a table?
TRUNCATE TABLE
29. What is the difference between DROP and TRUNCATE?
DROP deletes the table structure and data, while TRUNCATE removes all records but keeps the table structure.
30. How do you delete a view?
DROP VIEW Student_View;

D. Advanced Database Objects

31. How do you delete an index?

```
DROP INDEX idx_name
ON Student;
```

32. How do you execute a stored procedure?
CALL GetStudents();
33. How do you delete a stored procedure?
DROP PROCEDURE GetStudents;
34. What is a primary key?
A primary key uniquely identifies each record in a table.
35. What is a foreign key?
A foreign key establishes a relationship between two tables.
36. What is a constraint?
A constraint is a rule applied to table columns to maintain data accuracy and integrity.
37. Name some common SQL constraints.
PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and DEFAULT.
38. What is normalization?
Normalization is the process of organizing database tables to reduce data redundancy.
39. What is data integrity?
Data integrity ensures that data remains accurate and consistent.
40. Why should database objects be managed properly?
To maintain data accuracy, security, performance, and consistency.

E. Practical SQL Questions

41. How do you insert a record into a table?

```
INSERT INTO Student
VALUES (1, 'Rahim', 20);
```

42. How do you display all records?

```
SELECT * FROM Student;
```

43. How do you update a record?

```
UPDATE Student
SET Age = 21
WHERE ID = 1;
```

44. How do you delete a specific record?

```
DELETE FROM Student
WHERE ID = 1;
```

45. How do you display a view?

```
SELECT * FROM Student_View;
```

46. How do you check existing indexes?

```
SHOW INDEX FROM Student;
```

47. How do you display stored procedures?

```
SHOW PROCEDURE STATUS
WHERE Db = DATABASE();
```

48. How do you display triggers?

```
SHOW TRIGGERS;
```

49. How do you display functions?

```
SHOW FUNCTION STATUS
WHERE Db = DATABASE();
```

50. Which SQL command is used to remove a database?
DROP DATABASE

Short Questions and Answers Perform Data Manipulation Using SQL

1. What is data manipulation?
Data manipulation is the process of inserting, retrieving, updating, and deleting data in a database.
2. What is SQL?
SQL stands for Structured Query Language. It is used to manage and manipulate relational databases.
3. What is DML?
DML stands for Data Manipulation Language.
4. Name the common DML commands.
INSERT, UPDATE, DELETE, and SELECT.
5. What is the purpose of the SELECT command?
It retrieves data from a database table.
6. What is the purpose of the INSERT command?
It adds new records to a table.
7. What is the purpose of the UPDATE command?
It modifies existing records in a table.
8. What is the purpose of the DELETE command?
It removes records from a table.
9. What is a database record?
A record is a collection of related fields represented by a row in a table.
10. What is a database field?
A field is a column that stores a particular type of information.

B. SELECT Statement

11. Write the syntax of SELECT.
SELECT column_name
FROM table_name;
12. How do you retrieve all records from a table?
SELECT * FROM Student;
13. What is the WHERE clause?
WHERE is used to filter records based on specified conditions.
14. Write an SQL query to find students aged over 20.
SELECT * FROM Student
WHERE Age > 20;
15. What is the ORDER BY clause?
ORDER BY sorts records in ascending or descending order.
16. How do you sort student records by name?
SELECT * FROM Student
ORDER BY Name ASC;

17. What is DISTINCT?
DISTINCT removes duplicate values from query results.
18. Write a query to display unique departments.
SELECT DISTINCT Dept_ID
FROM Student;
19. What is the LIMIT clause?
LIMIT restricts the number of records returned by a query.
20. Write a query to display the first five students.
SELECT * FROM Student
LIMIT 5;

C. INSERT, UPDATE and DELETE

21. How do you insert a new record?
INSERT INTO Student (ID, Name, Age)
VALUES (1, 'Rahim', 20);
22. How do you insert multiple records?
INSERT INTO Student (ID, Name, Age)
VALUES
(2, 'Karim', 21),
(3, 'Hasan', 22);
23. How do you update a student's age?
UPDATE Student
SET Age = 23
WHERE ID = 1;
24. Why is WHERE important in UPDATE?
It ensures that only the specified records are updated.
25. How do you delete a specific record?
DELETE FROM Student
WHERE ID = 1;
26. What happens if DELETE is used without WHERE?
All records in the table are deleted, but the table structure remains.
27. What is the difference between DELETE and DROP?
DELETE removes records, while DROP removes the entire database object.
28. What is the difference between DELETE and TRUNCATE?
DELETE removes selected or all records; TRUNCATE removes all records while keeping the table structure.

D. SQL Operators and Functions

29. What is the AND operator?
AND returns records when all specified conditions are true.
30. What is the OR operator?
OR returns records when at least one specified condition is true.
31. What is the BETWEEN operator?
BETWEEN selects values within a specified range.
32. Write a query to find students aged between 18 and 25.
SELECT * FROM Student

WHERE Age BETWEEN 18 AND 25;
33. What is the IN operator?
IN checks whether a value matches any value in a specified list.
34. What is the LIKE operator?
LIKE searches for a specified pattern in a column.
35. What is the COUNT() function?
COUNT() returns the number of records.
36. What is the SUM() function?
SUM() calculates the total of numeric values.
37. What is the AVG() function?
AVG() calculates the average value.
38. What are MIN() and MAX() functions?
MIN() returns the smallest value, and MAX() returns the largest value.

E. Advanced SQL Queries

39. What is GROUP BY?
GROUP BY groups rows that have the same values in specified columns.
40. What is HAVING?
HAVING filters grouped records based on a condition.
41. What is the difference between WHERE and HAVING?
WHERE filters rows before grouping, while HAVING filters groups after grouping.
42. Write a query to count students in each department.

```
SELECT Dept_ID, COUNT(*) AS Total
FROM Student
GROUP BY Dept_ID;
```

43. What is a JOIN?
JOIN combines records from two or more tables based on a related column.
44. Name four common types of JOIN.
INNER JOIN, LEFT JOIN, RIGHT JOIN, and CROSS JOIN.
45. What is INNER JOIN?
INNER JOIN returns records that have matching values in both tables.
46. Write an INNER JOIN query.

```
SELECT Student.Name, Department.Dept_Name
FROM Student
INNER JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

47. What is a subquery?
A subquery is a query written inside another SQL query.
48. What is UNION?
UNION combines the results of two or more SELECT statements and removes duplicate rows.
49. What is a transaction?
A transaction is a group of database operations treated as a single logical unit.
50. What are COMMIT and ROLLBACK?
COMMIT saves transaction changes permanently, while ROLLBACK cancels uncommitted changes.

Short Questions and Answers on Apply Advanced SQL Techniques

A. Advanced SQL Concepts

1. What is advanced SQL?
Advanced SQL refers to techniques used to perform complex database operations and data analysis.
2. What is a subquery?
A subquery is a query written inside another SQL query.
3. What is a correlated subquery?
A correlated subquery is a subquery that depends on values from the outer query.
4. What is a JOIN?
JOIN combines data from two or more tables based on related columns.
5. Name the common types of JOIN.
INNER JOIN, LEFT JOIN, RIGHT JOIN, and CROSS JOIN.
6. What is SELF JOIN?
SELF JOIN joins a table with itself.
7. What is UNION?
UNION combines the results of multiple SELECT statements and removes duplicate rows.
8. What is UNION ALL?
UNION ALL combines query results and keeps duplicate rows.
9. What is the difference between UNION and UNION ALL?
UNION removes duplicates, while UNION ALL keeps duplicates.
10. What is a Common Table Expression (CTE)?
A CTE is a temporary named result set used within a single SQL statement.

B. Window Functions

11. What is a window function?
A window function performs calculations across related rows without combining them into one row.
12. What is the OVER() clause?
OVER() defines the set of rows used by a window function.
13. What is PARTITION BY?
PARTITION BY divides rows into groups for window function calculations.
14. What is ROW_NUMBER()?
ROW_NUMBER() assigns a unique sequential number to each row.
15. What is RANK()?
RANK() assigns ranking numbers and leaves gaps after tied values.
16. What is DENSE_RANK()?
DENSE_RANK() assigns ranking numbers without gaps after tied values.
17. Write a query using ROW_NUMBER().

```
SELECT Name, Marks,
ROW_NUMBER() OVER (ORDER BY Marks DESC) AS Row_No
FROM Student;
```

18. What is the difference between RANK() and DENSE_RANK()?
RANK() leaves gaps after ties, while DENSE_RANK() does not.

19. What is LAG()?
LAG() accesses a value from a previous row.
20. What is LEAD()?
LEAD() accesses a value from a following row.

C. Stored Procedures and Functions

21. What is a stored procedure?
A stored procedure is a collection of SQL statements stored in a database.
22. Which command executes a stored procedure in MySQL?
CALL
23. Write a command to execute a procedure.
CALL GetStudents();
24. What is a stored function?
A stored function is a database program that returns a value.
25. What is the difference between a procedure and a function?
A function must return a value, while a procedure does not have to return a value.
26. What is a parameter in a stored procedure?
A parameter is a value passed to a procedure or received from it.
27. What are IN, OUT, and INOUT parameters?
IN passes data into a procedure, OUT returns data, and INOUT passes and returns data.

D. Triggers and Transactions

28. What is a trigger?
A trigger is a database program that automatically executes when a specified event occurs.
29. What are the common MySQL trigger events?
INSERT, UPDATE, and DELETE.
30. What are BEFORE and AFTER triggers?
BEFORE triggers execute before an event, while AFTER triggers execute after an event.
31. What is a transaction?
A transaction is a group of database operations treated as one logical unit.
32. What is ACID?
ACID stands for Atomicity, Consistency, Isolation, and Durability.
33. What is COMMIT?
COMMIT permanently saves transaction changes.
34. What is ROLLBACK?
ROLLBACK cancels uncommitted transaction changes.
35. What is SAVEPOINT?
SAVEPOINT creates a point within a transaction to which we can roll back.
36. Write a transaction example.

```
START TRANSACTION;

UPDATE Account
SET Balance = Balance - 500
WHERE Account_ID = 1;

COMMIT;
```

E. Advanced Database Techniques

37. What is an index?

An index improves the speed of data retrieval from a table.

38. What is a composite index?

A composite index is an index created on two or more columns.

39. What is query optimization?

Query optimization is the process of improving SQL query performance.

40. What is EXPLAIN in MySQL?

EXPLAIN shows how MySQL executes a query.

41. Write an EXPLAIN command.

```
EXPLAIN
SELECT * FROM Student
WHERE Age > 20;
```

42. What is a view?

A view is a virtual table based on an SQL query.

43. What is a recursive CTE?

A recursive CTE is a CTE that refers to itself to process hierarchical or repeated data.

44. What is JSON data in MySQL?

JSON is a data format used to store structured information in JSON columns.

45. What is dynamic SQL?

Dynamic SQL is an SQL statement constructed and executed at runtime.

46. What is a cursor?

A cursor processes query result rows one at a time inside stored programs.

47. What is a temporary table?

A temporary table is a table that exists temporarily during a database session.

48. What is database partitioning?

Partitioning divides a large table into smaller manageable parts.

49. What is a deadlock?

A deadlock occurs when two or more transactions wait for each other to release locked resources.

50. What is SQL injection?

SQL injection is an attack in which malicious SQL code is inserted into application input.

Short Questions and Answers on Configure Database Security

1. Basic Database Security Questions

1. What is Database Security?
Answer: Database Security is the process of protecting a database from unauthorized access, misuse, modification, and data loss.
2. What is Configure Database Security?
Answer: Configure Database Security means setting up security rules, permissions, authentication, and protection mechanisms to secure a database.
3. Why is Database Security important?
Answer: It protects sensitive information and prevents unauthorized access, data theft, and data loss.
4. What is Authentication?
Answer: Authentication is the process of verifying the identity of a database user.
5. What is Authorization?
Answer: Authorization determines what actions an authenticated user is allowed to perform.
6. What is User Privilege?
Answer: User Privilege is permission granted to a user to perform specific database operations.
7. What is Access Control?
Answer: Access Control is a security mechanism that manages who can access database resources.
8. What is a Strong Password?
Answer: A strong password is a password containing uppercase letters, lowercase letters, numbers, and special characters.
9. What is Data Encryption?
Answer: Data Encryption is the process of converting readable data into an unreadable format to prevent unauthorized access.
10. What is SQL Injection?
Answer: SQL Injection is an attack that uses malicious SQL statements to access or manipulate database information.

2. Database Security Components

11. What is a Database User?
Answer: A Database User is an account that allows a person or application to access a database.
12. What is GRANT command?
Answer: GRANT is used to provide privileges to database users.
13. What is REVOKE command?
Answer: REVOKE is used to remove privileges from database users.
14. What is the use of CREATE USER command?
Answer: It is used to create a new database user account.
15. What is the use of DROP USER command?
Answer: It is used to remove a database user account.
16. What is Least Privilege Principle?
Answer: It means giving users only the minimum permissions required to perform their jobs.

17. What is a Database Firewall?
Answer: A Database Firewall monitors and filters database traffic to prevent unauthorized access and attacks.
18. What is an Audit Log?
Answer: An Audit Log records database activities, including user access and changes.
19. What is Database Backup?
Answer: Database Backup is the process of creating a copy of database information for recovery.
20. What is Database Recovery?
Answer: Database Recovery is the process of restoring lost or damaged database information.

3. MySQL Database Security Questions

21. How do you create a database user in MySQL?
Answer:

```
CREATE USER 'user1'@'localhost'
IDENTIFIED BY 'Strong@123';
```

22. How do you grant privileges to a user?
Answer:

```
GRANT SELECT, INSERT
ON college.*
TO 'user1'@'localhost';
```

23. How do you remove privileges from a user?
Answer:

```
REVOKE INSERT
ON college.*
FROM 'user1'@'localhost';
```

24. How do you display database users?
Answer:

```
SELECT User, Host FROM mysql.user;
```

25. How do you change a user's password?
Answer:

```
ALTER USER 'user1'@'localhost'
IDENTIFIED BY 'NewStrong@123';
```

26. How do you delete a database user?
Answer:

```
DROP USER 'user1'@'localhost';
```

27. How do you check user privileges?
Answer:

```
SHOW GRANTS FOR 'user1'@'localhost';
```

28. How do you grant all privileges to a user?
Answer:

```
GRANT ALL PRIVILEGES
ON college.*
TO 'user1'@'localhost';
```

29. What is the purpose of FLUSH PRIVILEGES?
Answer: It reloads the privilege tables. It is generally unnecessary after using GRANT or REVOKE in modern MySQL versions.
30. How can you prevent unauthorized database access?
Answer: Use strong passwords, user authentication, access permissions, encryption, firewall protection, and regular security updates.

4. Practical Competency Assessment Questions

No.	Practical Question	SQL Command
1	Create a database user.	CREATE USER
2	Grant SELECT permission.	GRANT SELECT
3	Grant INSERT permission.	GRANT INSERT
4	Revoke user permission.	REVOKE
5	Change user password.	ALTER USER
6	Check user privileges.	SHOW GRANTS
7	Delete a user account.	DROP USER
8	Display database users.	SELECT User, Host FROM mysql.user;
9	Back up a database.	mysqldump
10	Restore a database.	mysql command-line client

1. Basic Database Maintenance Questions

1. What is Database Maintenance?
Answer: Database Maintenance is the process of managing, updating, checking, and optimizing a database to ensure its proper performance.
2. Why is Database Maintenance important?
Answer: It improves database performance, ensures data accuracy, prevents data loss, and maintains security.
3. What is Database Backup?
Answer: Database Backup is the process of creating a copy of database data for future recovery.
4. What is Database Recovery?
Answer: Database Recovery is the process of restoring a database after data loss or failure.
5. What is Database Optimization?
Answer: Database Optimization is the process of improving database performance and query execution speed.
6. What is Database Monitoring?
Answer: Database Monitoring is the process of checking database performance, availability, and resource usage.
7. What is Database Restoration?
Answer: Database Restoration is the process of recovering database data from a backup file.
8. What is Data Integrity?
Answer: Data Integrity means maintaining the accuracy, consistency, and reliability of data.

9. What is Database Fragmentation?
Answer: Database Fragmentation occurs when data or indexes become inefficiently organized in storage.
10. What is Database Repair?
Answer: Database Repair is the process of fixing database table errors or corruption.

2. Database Maintenance Tools and Commands

11. What is the use of CHECK TABLE command?
Answer: It checks a table for errors.
12. What is the use of OPTIMIZE TABLE command?
Answer: It reorganizes table storage and can improve performance.
13. What is the use of REPAIR TABLE command?
Answer: It attempts to repair supported table types, such as MyISAM.
14. What is the use of ANALYZE TABLE command?
Answer: It updates table statistics used by the query optimizer.
15. What is the use of SHOW TABLES command?
Answer: It displays all tables in the selected database.
16. What is the use of DESCRIBE command?
Answer: It displays the structure of a table.
17. What is the use of SHOW DATABASES command?
Answer: It displays all available databases.
18. What is the use of DELETE command?
Answer: It removes selected records from a table.
19. What is the use of TRUNCATE command?
Answer: It removes all records from a table while keeping its structure.
20. What is the use of DROP command?
Answer: It permanently removes a database or table.

3. Database Backup and Recovery Questions

21. What is a Full Backup?
Answer: A Full Backup is a complete copy of the database.
22. What is an Incremental Backup?
Answer: It backs up data changed since the previous backup.
23. What is a Differential Backup?
Answer: It backs up data changed since the last full backup.
24. What is mysqldump?
Answer: mysqldump is a MySQL utility used to export and back up databases.
25. How do you take a MySQL database backup?
Answer: Use the following command:
mysqldump -u root -p college > college.sql
26. How do you restore a MySQL database?
Answer: Use the following command:
mysql -u root -p college < college.sql
27. What is a Backup Schedule?
Answer: It is a planned timetable for taking regular database backups.

28. Why should backups be stored in a separate location?
Answer: To protect data against hardware failure, accidental deletion, or disasters.

4. Database Performance and Security

29. What is Query Optimization?
Answer: Query Optimization is the process of improving SQL query execution efficiency.
30. What is an Index?
Answer: An Index is a database structure that helps retrieve data faster.
31. What is a Slow Query?
Answer: A Slow Query is an SQL query that takes a long time to execute.
32. What is EXPLAIN command?
Answer: EXPLAIN shows how MySQL plans to execute an SQL query.
33. What is Database Log?
Answer: A Database Log records database activities and events.
34. What is a Transaction Log?
Answer: It records transaction activities to support recovery and consistency.
35. What is Database Security Maintenance?
Answer: It involves updating passwords, checking permissions, monitoring access, and protecting database information.
36. What is Database Server?
Answer: A Database Server is a computer system or software that stores and manages databases.
37. What is Database Downtime?
Answer: Database Downtime is a period when the database is unavailable.
38. What is Database Migration?
Answer: Database Migration is the process of moving data from one database system to another.

1. Basic Cloud Database Questions

1. What is Cloud Database?
Answer: A Cloud Database is a database hosted on a cloud computing platform and accessed through the internet.
2. What is Cloud Computing?
Answer: Cloud Computing is the delivery of computing services over the internet.
3. What is Database Management on Cloud?
Answer: It is the process of creating, storing, managing, securing, and maintaining databases on cloud platforms.
4. What are the advantages of Cloud Database?
Answer: Scalability, flexibility, remote access, cost efficiency, and automatic backup.
5. What are the disadvantages of Cloud Database?
Answer: Internet dependency, security risks, service downtime, and ongoing subscription costs.

6. What is DBaaS?
Answer: DBaaS stands for Database as a Service. It provides database services through a cloud provider.
7. What is a Cloud Service Provider?
Answer: A Cloud Service Provider is a company that provides cloud computing services.
8. Name some popular Cloud Database Providers.
Answer: Amazon Web Services (AWS), Microsoft Azure, and Google Cloud.
9. What is a Public Cloud Database?
Answer: It is a database hosted on shared cloud infrastructure and provided over the internet.
10. What is a Private Cloud Database?
Answer: It is a database hosted on dedicated cloud infrastructure for a specific organization.

2. Cloud Database Services

11. What is Amazon RDS?
Answer: Amazon RDS is a managed relational database service provided by AWS.
12. What is Amazon Aurora?
Answer: Amazon Aurora is a cloud-native relational database service provided by AWS.
13. What is Google Cloud SQL?
Answer: Google Cloud SQL is a managed database service for MySQL, PostgreSQL, and SQL Server.
14. What is Azure SQL Database?
Answer: Azure SQL Database is a managed relational database service provided by Microsoft Azure.
15. What is MongoDB Atlas?
Answer: MongoDB Atlas is a cloud-based database service for MongoDB.
16. What is Firebase?
Answer: Firebase is a Google platform that provides cloud-based application development services, including database services.
17. What is Cloud Storage?
Answer: Cloud Storage is a service used to store data on remote cloud servers.
18. What is a Managed Database Service?
Answer: It is a database service where the cloud provider manages tasks such as maintenance, updates, and backups.

3. Cloud Database Configuration and Management

19. What is Cloud Database Configuration?
Answer: It is the process of setting up a database on a cloud platform.
20. What is Database Instance?
Answer: A Database Instance is an operating database environment running on a server.
21. What is Database Migration?
Answer: Database Migration is the process of transferring data from one database system to another.
22. What is Cloud Database Backup?
Answer: It is the process of creating a copy of cloud database data for recovery.
23. What is Cloud Database Restore?
Answer: It is the process of recovering cloud database data from a backup.

24. What is Database Replication?
Answer: Database Replication is the process of copying data from one database server to another.
25. What is Database Scaling?
Answer: Database Scaling means increasing or decreasing database resources according to workload.
26. What is Horizontal Scaling?
Answer: Horizontal Scaling means adding more database servers or nodes to handle workloads.
27. What is Vertical Scaling?
Answer: Vertical Scaling means increasing the CPU, RAM, or storage capacity of a database server.
28. What is High Availability?
Answer: High Availability ensures that a database remains accessible with minimal downtime.
29. What is Load Balancing?
Answer: Load Balancing distributes database requests across multiple servers or resources.
30. What is Disaster Recovery?
Answer: Disaster Recovery is the process of restoring database services after a major failure or disaster.

4. Cloud Database Security Questions

31. What is Cloud Database Security?
Answer: It is the process of protecting cloud databases from unauthorized access, attacks, and data loss.
32. What is IAM?
Answer: IAM stands for Identity and Access Management. It controls user identities and access permissions.
33. What is Multi-Factor Authentication (MFA)?
Answer: MFA is a security method that requires two or more verification factors to access an account.
34. What is Data Encryption?
Answer: Data Encryption converts readable information into an unreadable format to protect it.
35. What is SSL/TLS?
Answer: SSL/TLS provides encrypted communication between clients and database servers.
36. What is a Firewall?
Answer: A Firewall controls incoming and outgoing network traffic based on security rules.
37. What is Access Control?
Answer: Access Control determines which users can access database resources and perform operations.
38. What is a Security Group?
Answer: A Security Group acts as a virtual firewall that controls network traffic to cloud resources.

5. Cloud Database Monitoring and Maintenance

39. What is Cloud Database Monitoring?
Answer: It is the process of checking database performance, availability, and resource usage.

40. What is Database Performance?
Answer: Database Performance refers to how efficiently a database processes queries and manages data.
41. What is Auto Scaling?
Answer: Auto Scaling automatically adjusts computing resources according to workload.
42. What is Cloud Database Downtime?
Answer: It is the period when a cloud database service is unavailable.
43. What is a Database Log?
Answer: A Database Log records database activities, operations, and errors.
44. What is Point-in-Time Recovery (PITR)?
Answer: PITR restores a database to a specific point in time using available backups and transaction logs.
45. What is SLA?
Answer: SLA stands for Service Level Agreement. It defines the service commitments between a provider and a customer.

6. Practical Competency Assessment Questions

No.	Practical Question	Expected Task
1	Create a cloud account.	Register with a cloud provider.
2	Create a cloud database.	Configure a database instance.
3	Configure database settings.	Set database name and engine.
4	Connect to a cloud database.	Use a database client.
5	Create a database table.	Execute SQL commands.
6	Insert database records.	Use INSERT query.
7	Configure database security.	Set user permissions.
8	Configure a firewall.	Set access rules.
9	Take a database backup.	Create a backup copy.
10	Restore a database.	Recover data from backup.
11	Monitor database performance.	Check CPU, RAM and storage.
12	Configure database scaling.	Increase or decrease resources.